

P510/1

PHYSICS

PAPER 1

2HOURS 40 MINUTES



Apex Examination Council

PRE-MOCK EXAMINATION 2026 SET 1

UGANDA ADVANCED CERTIFICATE OF EDUCATION

PHYSICS

PAPER 1

TIME: 2 HOURS 40 MINUTES

INSTRUCTIONS TO CANDIDATES

- This paper consists of FOUR sections: A, B, C and D.
- Section A contains ONE compulsory item.
- Section B contains TWO items. Answer ONE item only.
- Section C contains TWO items. Answer ONE item only.
- Section D contains TWO items. Answer ONE item only.
- Answer FOUR items in all.
- Each item carries equal marks.
- Non-programmable calculators may be used.
- Credit will be given for clear working, correct units, relevant diagrams and logical presentation of answers.

Useful Constants

- Acceleration due to gravity, $g = 9.81 \text{ ms}^{-2}$
- Speed of light in vacuum, $c = 3.0 \times 10^8 \text{ ms}^{-1}$
- Planck's constant, $h = 6.63 \times 10^{-34} \text{ Js}$
- Electron charge, $e = 1.60 \times 10^{-19} \text{ C}$

SECTION A

Answer this item

ITEM 1

Uganda's Ministry of ICT and National Guidance has expanded digital learning programmes in schools. Learners now spend several hours each day using computers, tablets and smartphones equipped with LED screens.

Medical practitioners at a regional referral hospital have reported increasing cases of eye strain, headaches and sleep disturbances among learners. Researchers suspect that prolonged exposure to high-energy blue light emitted by LED screens may be responsible.

A team of Physics students was tasked with investigating whether photons emitted by blue light have sufficient energy to affect human eye tissues. The students obtained information showing that most LED screens emit blue light of wavelength 450 nm.

The Ministry intends to use the findings to educate the public on safe use of digital devices.

Support Materials

- Wavelength of blue light = 450 nm
- Planck's constant = 6.63×10^{-34} Js
- Speed of light = 3.0×10^8 ms⁻¹
- 1 eV = 1.60×10^{-19} J
- Retinal damage threshold = 3.1 eV

Tasks

- Calculate the energy of a photon emitted by the LED screen.
- Determine whether the photon energy exceeds the retinal damage threshold.
- Explain how quantum theory accounts for the emission of radiation from LED screens.
- Distinguish between ionising and non-ionising radiation.
- Explain the possible biological effects of prolonged exposure to high-energy radiation.
- Suggest measures that can minimise health risks associated with prolonged screen use

SECTION B

Answer one item only

ITEM 2

Following heavy rainfall in Bududa District, large masses of soil and rock slid down the slopes of Mount Elgon, destroying homes, roads and agricultural land.

Engineers investigating the disaster modelled a section of saturated soil as a block of mass 3000 kg resting on a slope inclined at 45° to the horizontal. Laboratory tests indicated that rainfall increased the soil mass by 25% due to water absorption. The coefficient of friction between the soil and underlying rock was estimated to be 0.45.

The Ministry of Works requires a scientific assessment of the stability of the slope.

Tasks

- a) Draw and label a force diagram showing all forces acting on the soil mass.
- b) Determine the component of the soil weight acting down the slope.
- c) Determine the normal reaction acting on the soil.
- d) Calculate the frictional force resisting motion.
- e) Assess whether the soil is likely to slide.
- f) Explain how rainfall, vegetation removal and slope angle influence landslides.
- g) Recommend measures for reducing future landslide disasters.

OR

ITEM 3

Traffic police in a mountainous district have reported frequent motorcycle accidents on a curved section of road during the rainy season.

Road engineers established that the bend has a radius of 12 m and is banked at an angle of 5° . The coefficient of friction between tyres and the road surface is 0.60 during dry conditions and 0.35 during wet conditions.

The combined mass of a motorcycle and rider is 180 kg.

Tasks

- Explain the concept of centripetal force.
- Determine the maximum safe speed on the bend under dry conditions.
- Determine the maximum safe speed under wet conditions.
- Explain the role of friction and banking in preventing skidding.
- Assess the dangers associated with excessive speed on curved roads.
- Suggest engineering measures that can improve road safety.

SECTION C

Answer one item only

ITEM 4

A rural health centre in Karamoja experiences frequent power outages, making it difficult to sterilise equipment and heat water required for maternity services.

Engineers installed a solar water-heating system consisting of a blackened absorber plate, insulated storage tank and reflective concentrators.

During evaluation of the system, students measured the following:

- Solar intensity = 800 Wm^{-2}
- Collector area = 0.20 m^2
- Efficiency = 35%
- Mass of water = 2 kg
- Initial temperature = 25°C
- Final temperature = 100°C
- Specific heat capacity of water = $4200 \text{ Jkg}^{-1}\text{K}^{-1}$

Tasks

- Explain the modes of heat transfer involved in the solar heater.
- Determine the thermal energy required to heat the water.
- Calculate the useful power supplied by the collector.

- d) Determine the time required to heat the water.
- e) Explain the energy transformations occurring in the system.
- f) Suggest improvements that can increase efficiency.

OR

ITEM 5

A remote health centre located in a valley experiences poor mobile network reception. Engineers propose the installation of a fibre optic communication system linking the centre to a nearby communication tower.

The optical fibre is made of glass with refractive index 1.50 while the surrounding medium has refractive index 1.00.

Tasks

- a) Explain the principle of total internal reflection.
- b) Determine the critical angle for the glass fibre.
- c) Explain how information is transmitted through optical fibres.
- d) Describe the advantages of fibre optic communication over metallic cables.
- e) Explain factors that may reduce signal quality.
- f) Suggest methods of improving communication reliability.

SECTION D

Answer one item only

ITEM 6

Several fire outbreaks have recently occurred in a busy urban market due to overloaded electrical installations.

An inspection revealed that a cable of length 20 m and resistance $0.5 \Omega \text{m}^{-1}$ was carrying a current of 8 A for approximately four hours each day.

Electrical engineers have been tasked with assessing the safety of the installation.

Tasks

- a) Determine the total resistance of the cable.
- b) Calculate the electrical power dissipated in the cable.
- c) Determine the energy dissipated during four hours of operation.
- d) Explain how overheating of conductors causes electrical fires.
- e) Describe the role of circuit breakers and fuses in electrical safety.
- f) Recommend measures that can reduce fire risks in markets.

OR

ITEM 7

A farming community intends to install a solar-powered electronic alarm system to detect and deter wild animals that destroy crops at night.

The system consists of a solar panel, battery, electromagnetic sensor, rectifier circuit and alarm unit. When an animal passes near the sensor, an electrical signal is generated and processed before activating the alarm.

Tasks

- a) Explain the principle of electromagnetic induction.
- b) Describe how the detection system operates.
- c) Draw and explain a full-wave rectifier circuit.
- d) Explain the function of a capacitor in the alarm system.
- e) Describe how a cathode ray oscilloscope (CRO) can be used to analyse the signal.
- f) Suggest improvements that can increase the efficiency and reliability of the system.

END